

Equations governing 2D discrete variable - Fourier Transform

2D Discrete Fourier Transform (DFT)

Given a 2D image signal $f(x, y)$ of size $M \times N$ (i.e., with $x=0, \dots, M-1$ and $y=0, \dots, N-1$), the **2D DFT** transforms it into a frequency domain signal $F(u, v)$, defined by:

Forward 2D DFT:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

Inverse 2D DFT:

To get back the image from its spectrum:

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) \cdot e^{j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

This brings you back from the frequency domain to the spatial image.

Here,

$f(x, y)$: represents original image intensity at pixel (x, y)

$F(u, v)$: represents complex frequency coefficient at (u, v)

u, v : frequency domain indices

j : imaginary unit $\sqrt{-1}$

Real-World Interpretation

Low frequencies $(u, v) \approx (0, 0)$: capture smooth changes, general shapes.

High frequencies (u, v) far from zero: capture edges, fine details, noise.